

The Universal Bounding Theorem of Coherent Residual Pathways: State Independence and Observable Symmetries Extension 12

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Abstract

We formulate the ultimate generalization of the Q-LINK architecture by lifting the constraints on the initial state preparation and the observable basis. By evaluating the pure geometric properties of the density matrix and the Pauli trace inner products, we prove the ‘‘Purity Axiom,’’ establishing that the architectural advantage is completely invariant under local initial state transformations but degrades catastrophically under mixed states. Furthermore, we generalize the combinatorial variance shield to arbitrary m -weight non-diagonal observables, and establish the bounds of Hamiltonian symmetry fracturing. This defines the absolute, most general boundary of the residual ansatz.

1 The Purity Axiom (Initial State Independence)

The variational cost function evaluates the expectation of an observable: $\mathcal{L}(\theta) = \text{Tr}(\rho_{tot} U^\dagger(\theta) \tilde{O} U(\theta))$. The global initial state is $\rho_{tot} = \rho_0 \otimes \rho_m$.

According to the Master Formula, the absolute gradient variance scales directly with the squared Hilbert-Schmidt norm of the traceless initial state $H = \rho_{tot} - \frac{1}{d}I$:

$$\text{Tr}(H^2) = \text{Tr}(\rho_{tot}^2) - \frac{1}{d} \quad (1)$$

Notice that equation (1) depends strictly on the *purity* $P = \text{Tr}(\rho_{tot}^2)$ of the initial state, not its vector coordinates.

Theorem of State Independence: If ρ_{tot} is a pure state ($P = 1$), the ensemble variance achieves its theoretical maximum $1 - \frac{1}{d}$. The choice of $|0\rangle^{\otimes n}$ is merely a computational convention. Any unitary state preparation V_{prep} applied before the variational ansatz (e.g., creating superpositions or entangled states) commutes through the Haar integration volume and leaves the asymptotic variance bound strictly invariant.

The Mixed State Penalty: If the state is mixed (e.g., due to noise or thermal initialization), $P < 1$, and the trace $\text{Tr}(H^2)$ shrinks linearly. The structural advantage of the Q-LINK architecture cannot rescue a gradient signal destroyed by initial state decoherence.

2 The Basis-Blind Combinatorial Shield

In previous extensions, the m -local cost operator was defined exclusively using the Z -basis. We now generalize this to an arbitrary observable.

Let the cost operator on the data register be an arbitrary superposition of Pauli strings P_j :

$$O_{data} = \sum_j c_j P_j \quad (2)$$

The Hilbert-Schmidt trace inner product of any two Pauli strings satisfies $\text{Tr}(P_i P_j) = d\delta_{ij}$. Because the trace norm depends only on the orthogonality of the Pauli group, the specific basis (X, Y , or Z) of the operators is geometrically irrelevant.

Generalized Combinatorial Bound: Let O_m be any normalized operator constructed from Pauli strings strictly of weight m acting on the n data qubits. The traceless part $\tilde{V}_m = O_m \otimes I_m$ evaluates to:

$$\text{Tr}(\tilde{V}_m^2) = \frac{2N}{\binom{n}{m}} \quad (3)$$

The asymptotic variance ratio $\mathcal{R}_m(n) = \frac{n+1}{n+1-m}$ applies universally to highly entangled, non-diagonal Hamiltonians (e.g., Transverse Field Ising Models or arbitrary fermion-mapped molecular Hamiltonians), provided the measurement support is strictly bounded to the data register.

3 Hamiltonian Symmetry Fracturing

The final theoretical limit emerges when the generalized cost operator O_{data} possesses a global symmetry that commutes with the Dynamical Lie Algebra (DLA) of the Q-LINK circuit.

Let $\mathcal{S}$ be a symmetry operator such that $[\mathcal{S}, O_{data}] = 0$ and $[\mathcal{S}, \mathfrak{g}_A] = 0$. By Schur's Lemma, the extended Hilbert space decomposes into invariant subspaces (sectors) labeled by the eigenvalues of $\mathcal{S}$:

$$\mathcal{H}_{tot} = \bigoplus_s \mathcal{H}_s \quad (4)$$

Because the Q-LINK generators cannot cross these sectors, the Haar measure does not integrate over $SU(2^{n+1})$, but rather over the direct product of local unitary groups $\bigotimes SU(d_s)$.

Theorem of Symmetry Bounds: If the observable shatters the DLA, the Q-LINK trace advantage must be evaluated strictly within the populated symmetry sector $\mathcal{H}_s$ defined by the initial state. The combinatorial shield equation $\mathcal{R}_m(n)$ must be mapped onto the localized dimension d_s , which fundamentally alters the algebraic ratio, often violently suppressing the barren plateau dynamically without requiring structural architectural changes.

4 The Absolute Generalization

The generalized form of the paper dictates that the mitigation of barren plateaus via coherent residual pathways is governed strictly by three geometric pillars: 1. The dimensional asymmetry of the measurement support ($\mathcal{H}_{data}$ vs $\mathcal{H}_{tot}$). 2. The anti-commutativity of the messenger's local Lie generators (The Orthogonal Hinge). 3. The intersection of the observable's support weight (m) and its symmetry invariants ($\mathcal{S}$).

All other parameters—initial states, operational unitary bases, temporal frequencies, and specific Pauli measurements—are fully elastic hyperparameters that navigate the finite-depth landscape but do not alter the fundamental asymptotic bounding box.